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Instructions: You might use rough sheets to work out your answers but the final answers must be written in the space provided on this question paper.

[Question 1] [Local Search and Perceptron Learning]

[5 + 5 + 5 Points]

Part a) The n-queens problem is about finding a way to place n queens on a an n x n chess board such that none Queens attack each other

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In this problem you are required to use local search to find optimal solution of 3 queens' problem. For this problem, assume that a solution is represented as an array of 3 non-negative integers in the range 0-2 with **Kth** element of the array indicating the row number of the **Kth** queen in the **Kth** column of the board. For example the solution shown above will be written as [1 2 0] hence the first queen is in row number 1 column number 0 the second is in row 2 column 1 and the third is in row 0 column 2.

Assume that the neighbors of a solution are found/generated by changing row number of a Queen so 6 different solutions are neighbor of each solution.

Use local search to find optimal solution of 3 queens problem starting with the randomly generated solution [1 2 1]. Clearly show each neighbor used during each iteration of the local search.

Randomly generated solution = [1, 2, 1] (3 attacking pairs)

Neighbours = [0, 2, 1] (1 attacking pairs)
 [2, 2, 1] (2 attacking pairs)
 [1, 0, 1] (3 attacking pairs)
 [1, 1, 1] (3 attacking pairs)
 [1, 2, 0] (1 attacking pair)
 [1, 2, 2] (2 attacking pairs)

2nd Iteration :-

Selected Solution = $[0, 2, 1]$ (1 attacking pair)

Neighbours = $[1, 2, 1]$ (3 attacking pairs)

$[2, 2, 1]$ (2 attacking pairs)

$[0, 0, 1]$ (2 attacking pairs)

$[0, 1, 1]$ (2 attacking pairs)

$[0, 2, 0]$ (1 attacking pair)

$[0, 2, 2]$ (2 attacking pairs)

As, no new better solution was found, hence the solution stops and the final answer is $[0, 2, 1]$ with only 1 attacking pair.

Part b) Subset-sum is an important NP-complete decision problem in computer science and can be stated as follows:

Given a multi-set of integers, is there a non-empty subset whose sum is exactly equal to a given value S .

For example, given the multi-set $\{-7, -7, -3, -2, 9000, 5, 8\}$ of seven integers and the value -19 , the answer to the subset sum problem is **YES** because the multi-subset $\{-7, -7, -3, -2\}$ sums to -19 . Similarly for the given set $\{-7, -3, -2, 9000, 5, 8\}$ of six integers and the value -5 , the answer to the subset sum problem is **YES** because several subset exists that sums to -5 like $\{-3, -2\}$ and $\{-7, -3, 5\}$ etc.

Furthermore for the same set $\{-7, -3, -2, 9000, 5, 8\}$ of six integers and the value 18000 , the answer to the subset sum problem is **NO** because no subset exists that sums to 18000 .

In this question we are going to pose the subset sum problem of described in first question as an optimization problem and then use Hill climbing strategy (i.e. a local search algorithm) to solve it.

Assume that for a set having n elements, a solution is coded using a bit string of length n with a bit being set to 1 if the element is part of the subset and 0 otherwise. Further, assume that the optimality of a solution is computed using $1 / (|S - \Sigma| + 1)$ where S is the required value of sum and Σ is sum of the subset and $|x|$ represents absolute value of x .

A simple operator to generate a new solution from an existing solution can be defined as follows

NEW_SOLUTION(X) = FLIP A BIT IN THE SOLUTION X

This is equivalent to including an element in the subset or excluding an already chosen element from the subset. It is obvious that for a set of size n we can generate n new solutions from an existing solution.

Use the above operator for generating new solutions along with the hill climbing search strategy (also known as local search) to find a solution for the following subset sum problem.

Find a subset of the set $\{2, 3, 4, 8, 16\}$ having sum 17. Take the solution 00000 as the starting solution in your local search. You must show all intermediate steps in the form of the following table. For each iteration show all intermediate solutions considered/generated and the solution selected at that iteration.

Iteration 1

Neighbours =
 $10000 \rightarrow \text{Fitness} = (1 / (|17 - 2| + 1)) = 1/16$
 $01000 \rightarrow \text{Fitness} = (1 / (|17 - 3| + 1)) = 1/15$
 $00100 \rightarrow \text{Fitness} = (1 / (|17 - 4| + 1)) = 1/14$
 $00010 \rightarrow \text{Fitness} = (1 / (|17 - 8| + 1)) = 1/10$
 $00001 \rightarrow \text{Fitness} = (1 / (|17 - 16| + 1)) = 1/2$

2nd Iteration

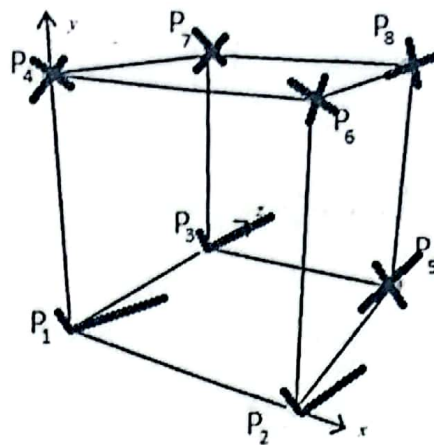
Selected Solution = ~~00000~~ 00001 (Fitness = ~~1/2~~ 9.5 ; highest)

Neighbours =
 $10001 \rightarrow \text{Fitness} = 1/2$
 $01001 \rightarrow \text{Fitness} = 1/3$
 $00101 \rightarrow \text{Fitness} = 1/4$
 $00011 \rightarrow \text{Fitness} = 1/8$
 $00000 \rightarrow \text{Fitness} = 1/17$

As, no better solution found, hence stop final solution is 00001 that is subset $\{16\}$.

(1, 0, 1) respectively

the coordinate plane i.e. (0, 0, 0) and points P_4 and P_6 have coordinates (0, 1, 1) and



Using a single epoch of perceptron learning rule to learn a linear model for this data that. You must start with the initial weights $[-1 \ 0 \ 0 \ 0]$ and use the points in order i.e. P_1 then $P_2 \dots P_8$ during the learning process. Further, the first weight i.e. -1 is that of the bias term where the bias term is always 1 and learning rate is also 0.5.

Show all the working that you did to calculate the values of the weights. The activation function $g(x)$ is given below:

$$g(x) = \begin{cases} 1, & x \geq 0 \\ -1, & x < 0 \end{cases}$$

Iteration No 1

With weights $[-1 \ 0 \ 0 \ 0]$

Example	Computed output	Weight
$P_1(0,0,0)$	$\eta = -1 \times 1 + 0 \times 0 + 0 \times 0 + 0 \times 0 = -1$ and hence $g(\eta) = -1$ INCORRECT	NO CHANGE IN WEIGHTS NEW WEIGHTS $[-1 \ 0 \ 0 \ 0] - [0.5 \ 0 \ 0 \ 0]$ $= [-1.5 \ 0 \ 0 \ 0]$
$P_2(1,0,0)$	$\eta = -1.5 \times 1 + 0 \times 1 + 0 \times 0 + 0 \times 0 = -1.5$ and hence $g(\eta) = -1$ INCORRECT	NEW WEIGHTS $[-1.5 \ 0 \ 0 \ 0] - [0.5 \ 0.5 \ 0 \ 0]$ $= [-2 \ -0.5 \ 0 \ 0]$
$P_3(0,0,1)$	$\eta = -2 \times 1 + -0.5 \times 0 + 0 \times 0 + 0 \times 1 = -2$ and hence $g(\eta) = -1$ INCORRECT	NEW WEIGHTS $[-2 \ -0.5 \ 0 \ 0] - [0.5 \ 0 \ 0 \ 0.5]$ $= [-2.5 \ -0.5 \ 0 \ -0.5]$
$P_4(0,1,0)$	$\eta = -2.5 \times 1 + -0.5 \times 0 + 0 \times 1 + -0.5 \times 0 = -2.5$ and hence $g(\eta) = -1$ CORRECT	NO CHANGE IN WEIGHTS

$\square$ and $\square$ $P5(1,0,1)$	$n = -2.5 \times 1 + -0.5 \times 1 + 0 \times 0 + -0.5 \times 1 = -3.5$ and hence $g(n) = -1$ CORRECT NO CHANGE IN WEIGHTS
$P6(1,1,0)$	$n = -2.5 \times 1 + -0.5 \times 1 + 0 \times 1 + -0.5 \times 0$ $n = -3.5$ and hence $g(n) = -1$ CORRECT NO CHANGE IN WEIGHTS
$P7(0,1,1)$	$n = -2.5 \times 1 + -0.5 \times 0 + 0 \times 1 + -0.5 \times 1 = -3$ and hence $g(n) = -1$ CORRECT NO CHANGE IN WEIGHTS
$P8(1,1,1)$	$n = -2.5 \times 1 + -0.5 \times 1 + 0 \times 1 + -0.5 \times 1 = -3.5$ and hence $g(n) = -1$ CORRECT NO CHANGE IN WEIGHTS

Final weights = $[-2.5, -0.5, 0, -0.5]$